

MAJOR PROJECT REGISTRATION FORM

Date: 31 - 07 - 2026

Project Category:	☒ Research oriented	☐ Application oriented	Batch ID:	AIML – C16
	☐ Product oriented Specify if others: —	☐ Other (please specify)		

Title of the Project: Explainable AI-Based Brain Tumor Detection using an Enhanced CNN Architecture

Abstract

Brain tumor detection from Magnetic Resonance Imaging (MRI) scans plays a vital role in early diagnosis and effective clinical decision-making. Although Convolutional Neural Networks (CNNs) have demonstrated significant potential for automated brain tumor classification, conventional architectures are often affected by overfitting, limited attention to tumor-specific regions, and poor interpretability. To address these limitations, this research proposes an Explainable Artificial Intelligence (XAI)-based brain tumor detection framework employing an enhanced CNN architecture that integrates Batch Normalization, the Convolutional Block Attention Module (CBAM), and Dropout. Batch Normalization improves training stability and convergence, while CBAM enables the network to emphasize spatially and channel-wise relevant features associated with tumor regions. Dropout is incorporated as a regularization technique to minimize overfitting and improve the model's generalization capability. Furthermore, Grad-CAM (Gradient-weighted Class Activation Mapping) is employed to enhance model interpretability by generating visual heatmaps that identify the regions of MRI scans contributing to the model's predictions. The proposed framework classifies MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. Its effectiveness is evaluated against a baseline CNN model using performance metrics including accuracy, precision, recall, F1-score, validation loss, and training time. The proposed approach aims to improve classification performance while providing transparent and interpretable predictions. By integrating enhanced deep learning techniques with Explainable AI, the framework offers a reliable and interpretable approach to brain tumor detection that can support medical professionals in diagnostic assessment and decision-making.

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